Start here for students and parents

You choose one route: Machine Learning or Python for Data Analysis. Both start from the beginning. The camp has no tests and no marks. In every class you build something and save it, and at the end you show a project to your family. S.4–S.6 students and vacists are welcome; you do not need coding experience or a particular subject combination.

Before the first class

The camp is held at The National ICT Innovation Hub, Nakawa. Some learners take one class a week in person on Saturday and the other classes online; the organiser tells you which pattern your group follows. Dates and times are confirmed with each group before the first class, and parents and guardians get the class times in writing. Read HYBRID_DELIVERY.pdf for the weekly pattern, the online checklist and the safety rules.

Every learner must have their own laptop and bring its charger, a notebook and a pen. A phone is not sufficient. Arrange a device check before paying. Use a supported computer system and do not install software from links in messages from strangers. The instructor will help with the official tools and check that you can open the first lesson. Tell the instructor about access needs or a device problem early. You do not need a public Google, GitHub or AI account to complete these materials.

Open your student pack

1. Download the ZIP for your chosen track. A ZIP is a compressed folder of files.
2. On Windows, right-click it and choose Extract All. On a Mac, double-click it. Work in the extracted folder, not inside the ZIP viewer.
3. Move the extracted folder to Documents and give it a simple name, such as HolidayCamp. Keep all supplied data files beside your lesson files.
4. Print My_Progress_Map.pdf (one A4 page) or ask the instructor for a printed copy. Open START_HERE.pdf with a text editor, or ask the instructor for the printed copy. ML01/DA01 is the first lesson; the numbers are the teaching order. Do not attempt the lessons alone in advance. ML adds F01 after ML03, F02 after ML06, F03 after ML09 and F04 after ML12; see FOUNDATION_LABS.pdf. These are taught labs on a fourth day in weeks 1–4. Both tracks otherwise have three classes per week.5. Keep a separate folder named MyWork for exported reports and hand-in copies. Run the supplied notebooks beside their CSV files; if you move a notebook into MyWork, copy its data files there too. Use a nickname or an organiser-issued learner code in filenames. Do not use a university registration number. Do not put private information into sample datasets.

Python and Jupyter for both tracks

The instructor prepares Python and Jupyter Notebook before class. Python runs instructions; Jupyter displays notebook pages. A file ending .ipynb is a notebook. It is not a Word document. Open Jupyter from the prepared shortcut or launcher, use its file list to enter your extracted camp folder, and select DA01.ipynb for Data Analysis or ML01.ipynb for Machine Learning. Click a code cell and press Shift+Enter to run it. A text cell explains; a code cell executes. Save with Ctrl+S on Windows or Command+S on Mac. A browser tab showing a local notebook does not mean your file has been shared with the instructor.

If you see text containing braces instead of a notebook page, you opened the file in a text editor; return to Jupyter. If a lesson says a package is missing, ask for the prepared environment instead of pasting unknown installation commands. Never share a notebook-server token or password.

The instructor shows how to restart the kernel (the running Python process) and run all cells from top to bottom. This exposes missing steps hidden in memory. Save a fresh copy before experimenting. The .md lesson sheet repeats the task in a form you can read without running code.

Your tools

Your tools are Python, NumPy, pandas, matplotlib, seaborn and scikit-learn, all in Jupyter Notebook. They are free. The first six classes teach Python from the beginning, pandas starts in class seven, and your first prediction is in class one. The four foundation labs use NumPy. scikit-learn arrives in class 13. You do not need any account or paid software.

Your data is camp_shop.csv (a 60-row practice shop), synthetic_homes.csv, synthetic_fares.csv and the iris flowers that come with scikit-learn. The data is invented, so it says nothing about real homes, trips or shops. Keep the CSV files beside your notebooks.

Your final project folder includes project.ipynb, your data dictionary, report.pdf or report.docx and AI_USE_LOG.md. Save five presentation slides as showcase.pptx for the parent showcase.

How each class works

Every class follows the same steps. Read the words for today. Predict what a small example will do and write your guess on paper. Run it and compare. Investigate by answering two or three questions about the output. Modify one value and explain the new result. Then do the independent Make task, save it, and add a line about it to your build log. Each sheet ends with an error card for the two errors most likely that day and an exit ticket sentence. Ask whenever a word, symbol or instruction is unclear. The instructor checks each learner's work in the small group.

Each sheet starts with a "Where this is used" box. It tells you where the day's idea shows up in real life, and asks one question to think about.

Online classes follow the same steps. In the 90 minutes live you predict, run and investigate with the instructor. The Modify steps and the Make task become your build task, about an hour in your own time. You send a photo of your work or the notebook by the route the organiser sets, before the next class. HYBRID_DELIVERY.pdf has the details.

Use two 45 to 60 minute practice blocks on non-class days as a starting routine. Take breaks. If stuck for 30 minutes, save the error and the question for class instead of copying an answer you cannot explain. Practice is expected; it is how the skills become usable.

Your progress map and your project

My_Progress_Map.pdf shows the eight weeks of the camp. Each week has a badge, a product you will build and three stamps, one for each class (plus a small stamp for each foundation lab in weeks 1 to 4). Colour in a stamp when you finish a class and show your build, or ask the instructor to stamp it. Three stamps win the week's badge. The finish line is the parent showcase.

In class 1, pick a topic you love, such as football, music, farming or fashion, and write it in the "My project" box on your progress map. That topic is the start of your own project. You write one idea for it in your build log after most classes. In week 4 you turn the topic into a project question, and in week 7 you turn the question into a plan. The supplied data files are the safe default. Real public data can be used only after you and the instructor have read its licence.

How to share your work and keep your build log

Keep a build log: one short entry for every class, with the file you saved, one sentence on what you made and one sentence on what to try next. You can add a fifth line, "Project corner": one idea for your project. BUILD_LOG_AND_SHOWCASE.pdf has an example. The organiser will issue a private route for sharing your project folder and the actual dates. These materials do not contain a live school upload form. Use only the route named for this camp. Do not use an IUEA university assignment form.

1. Read the lesson's filenames and formats. For your project, include the complete set of files.
2. Save, close and reopen each file. Check that code runs, Python notebooks rerun and report text can be selected.
3. In Word use Save As for DOCX or Export for PDF. In LibreOffice Writer use Save As for DOCX or Export as PDF. For presentations use PowerPoint or Impress and save PPTX. Keep editable originals. Do not substitute photos of pages for an editable or text-readable file.
4. Open the issued collection link or folder, sign in only if the organiser's chosen route requires it, and upload the files. If a form is used, finish its last step. Keep the confirmation.
5. Keep a local copy. If uploading fails, preserve your files and contact the organiser through the agreed channel. Saving locally is not the same as sharing.

Projects and certificates

You will show your own project to parents and guardians at the showcase in the final week. You can share a local or private copy. Public posting is optional and needs agreement. You will also have friendly show-and-tells along the way. Your instructor gives you one strength and one next step, kindly and privately. You earn the programme certificate by attending or making up most classes, keeping your build log and showing a finished project. There are no marks. The certificate is a programme certificate. It is not a university qualification or a certificate from Google, IBM or another course provider.

Make your first short presentation

Open PowerPoint or LibreOffice Impress and choose a blank presentation. A slide is one screen of information. Click the title box and type your project question. Add a new slide using Home > New Slide in PowerPoint or Slide > New Slide in Impress. Use a title and content layout for one main idea at a time. Insert your saved chart or screenshot using Insert > Pictures/Image; do not stretch it so labels become distorted. Use the Notes area below the slide to write what you will say, rather than placing your whole speech on the slide.

Create the five slides specified in your lesson. Save as PPTX, close it and reopen it. Start the slide show from the beginning and practise with a five-minute timer. Keep a PDF or screenshots as a backup. Check that no names, contact details or private notes appear in a presentation image. Menu names may differ slightly across versions; ask the instructor to show the installed version before starting.

Responsible AI literacy

Read AI_USE_POLICY.pdf. Weeks 1–6: no AI-generated solutions for core exercises, labs or build tasks. Weeks 7–8: demonstration, critique and limited supervised use only after independent foundations; keep AI_USE_LOG.md for any permitted assistance. Students must understand, test and explain all work. The showcase is presented without an AI assistant open. Core exercises stay free of AI-generated solutions throughout. Session 21 uses AI_CODING_ASSISTANTS_LAB.pdf within the existing practice time.